

knowledge [57, 42]. In these settings, items like movies, people, or products are modeled as nodes (for example, a particular film, an actor, or a smartphone), while their properties and connections (such as “directed by”, “co-acted with”, “has price”, or “manufactured by”) are represented as labeled edges. This simple pattern—a set of comparable items, each described along a common set of properties—underlies many user-facing services: one can quickly retrieve “all films directed by X after 2010” or “all products with feature Y below price Z” because the relevant facts are stored in a consistent, machine-readable structure rather than buried in free text.

A similar principle underlies the Open Research Knowledge Graph (ORKG) [6, 7], an online platform (<https://orkg.org/>) that structures scientific literature in the same way: research papers, materials, processes, and experimental conditions are treated as entities, and their relationships are captured through explicitly defined properties. Instead of static PDF tables—in which column headings such as “matrix”, “annealing temperature”, or “threshold voltage” may recur across papers but remain disconnected and non-interoperable—the ORKG models these aspects as explicit, searchable properties. Relations such as “precursor”, “co-reactant”, or “growth per cycle” become named predicates that can be aligned and reused across works [16, 26].

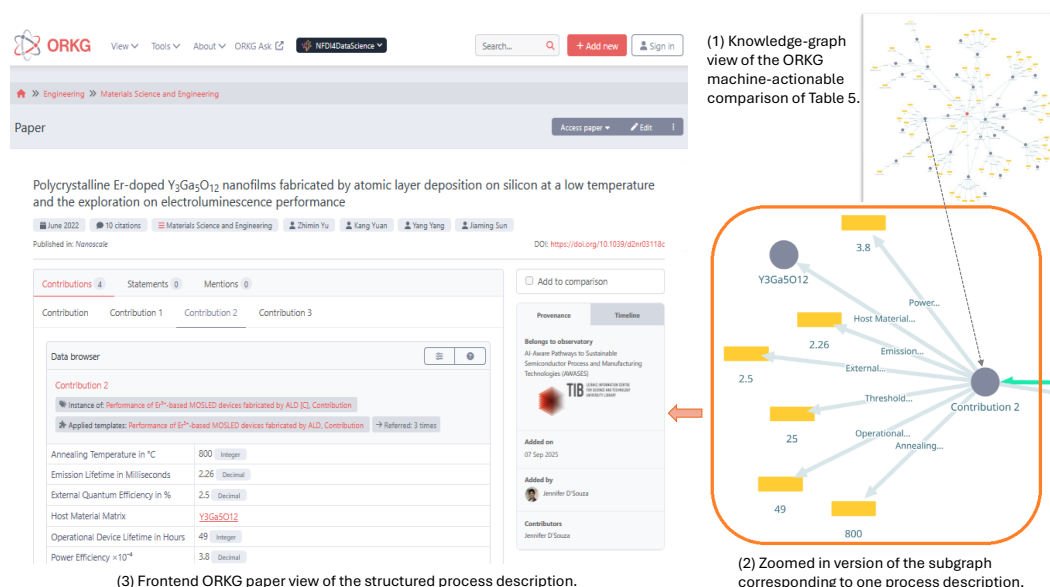


Figure 2: Continuation of Figure 1. Starts with Panel (1) which is the knowledge graph view (the actual graph representation of the machine-actionable Table 5 [25]). Panel 2 zooms in on the first contribution, linking the comparison entry to its individual paper record in the ORKG <https://orkg.org/papers/R1469778/R1469970> (Panel 3), which corresponds to the first data row in Table 5 in the review by Ghazy et al [25] and the article “*Polycrystalline Er-doped $Y_3Ga_5O_{12}$ nanofilms fabricated by atomic layer deposition on silicon at a low temperature and the exploration on electroluminescence performance*” [68].

In practice, a conventional survey or review table is decomposed so that each *row* becomes a separate *contribution* (one experimental configuration or reported device), and each *column* becomes a well-defined property. Figure 1 and 2 illustrates this for Table 5 in the rare-earth ALD review by Ghazy et al. (2025) [25]. Panel (1) reproduces the original table; Panel (2) shows the corresponding ORKG comparison (<https://orkg.org/comparisons/R1469991>), where the nine table rows become nine contributions and the table headers appear as ORKG properties (gray column).

Each of these properties corresponds to a performance descriptor routinely interpreted by ALD experimentalists:

- **Host material matrix** (P180031): the crystalline or amorphous material system into which Er^{3+} is incorporated (e.g., $Y_3Ga_5O_{12}$, Ga_2O_3).
- **Annealing temperature** (P180032): the post-deposition thermal treatment used to activate dopants, control defects, or enhance crystallinity.

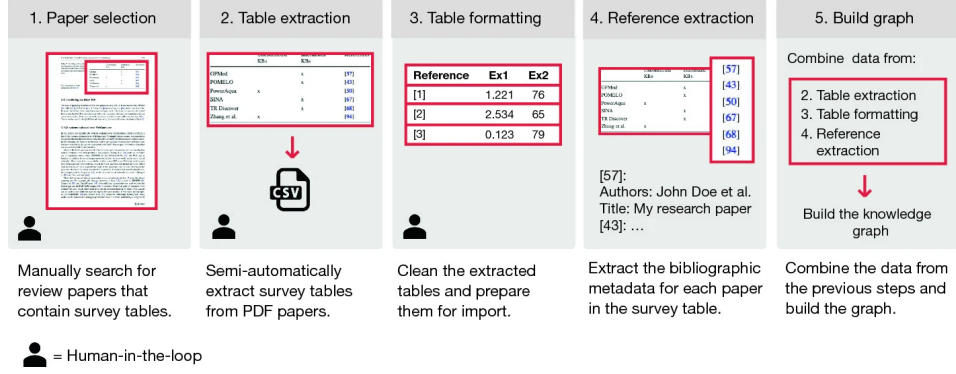


Figure 3: Methodology for importing survey tables into the Open Research Knowledge Graph (ORKG). Figure reproduced from Oelen *et al.* (2020), “Creating a Scholarly Knowledge Graph from Survey Article Tables” [45], where this technique was first introduced.

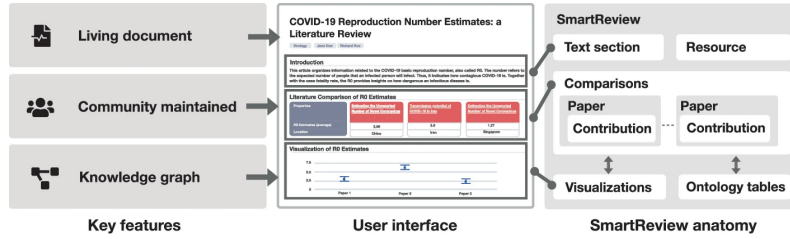


Figure 4: Illustration of key features and anatomy of SmartReviews. They are composed of several building blocks, including natural text, comparisons, and visualizations. Figure reproduced from Oelen *et al.* (2021), “SmartReviews: Towards Human- and Machine-Actionable Representation of Review Articles” [46], where the SmartReview methodology was first introduced.

- **External quantum efficiency (EQE)** (P180033): the fraction of injected charge carriers converted into emitted photons.
- **Power efficiency (PE)** (P180034): the optical output power per unit electrical input, here reported as values scaled by $\times 10^{-4}$.
- **Threshold voltage** (P180035): the voltage at which electroluminescence begins.
- **Emission lifetime** (P180036): the decay time of the 1530–1550 nm Er^{3+} emission, indicating radiative efficiency and defect environments.
- **Operational device lifetime (OLT)** (P180037): the duration for which the MOSLED device maintains usable output under operation.

Encoded as explicit predicates, these seven descriptors transform a single PDF table into a small, searchable performance dataset for Er^{3+} -based MOSLED devices. For the ALD community, the ORKG still behaves like a familiar comparison table, but—as a knowledge-graph representation—it can also be searched, filtered, linked to additional ALD/E data, and programmatically reused in ways not possible with static PDF tables.

For the ORKG platform, Oelen *et al.* (2020) first introduced *ORKG Comparisons* as a dedicated content type for representing and comparing research contributions [44]. Conceptually, an ORKG Comparison is a structured comparison table in which:

- each column is a *contribution*, representing one reported device or experimental configuration from a specific research article (e.g. one of the nine Er^{3+} -based MOSLED devices listed in Table 5 of Ghazy *et al.* (2025)). Each contribution is linked directly to the ORKG entry of its source publication, ensuring full traceability so that the underlying article can be opened, inspected, and verified;
- each row is a *typed property*: numerical quantities such as annealing temperature, external quantum efficiency (EQE), power efficiency, threshold voltage, emission lifetime, and

operational device lifetime (OLT) are stored as numbers with units, while categorical aspects such as the host material matrix are stored as controlled values rather than arbitrary free text;

- each cell can hold a *resource-like value*, meaning that recurring entities (for example, a specific host material matrix such as $\text{Y}_3\text{Ga}_5\text{O}_{12}$) are represented once as an identifiable resource (e.g. <https://orkg.org/resources/R1469904>) and reused across many contributions and comparisons;
- the comparison remains a *living* object that can be updated or extended when new data or corrections become available (e.g. adding further Er^{3+} -based MOSLED devices as the literature grows);
- recurring column sets can be captured as *templates*, making it easy to apply the same property schema (in this example: host material matrix, annealing temperature, EQE, power efficiency, threshold voltage, emission lifetime, and OLT) across multiple review tables.

In a follow-up study, Oelen *et al.* (2020) demonstrated how to populate such comparisons from already published survey tables [45]. As illustrated in Figure 3, the workflow starts from PDF tables, extracts the cell contents, aligns the extracted columns to a simple schema (a description of what each column means), and then lets curators check and refine the result. Tables that already contain a dedicated “Ref.” or “Reference” column are particularly suitable: each row naturally corresponds to a contribution linked to a specific paper, and each column header becomes a candidate ORKG property. The outcome is *FAIR* scientific data (findable, accessible, interoperable, reusable) [62]: tables are no longer just images in a PDF, but machine-readable records that can be searched, filtered, and recombined in new analyses. While the survey-table setting provides a direct mapping from rows and columns to ORKG Comparisons, the same idea extends to other kinds of scientific knowledge (e.g. text summaries, figure panels, or protocol descriptions). In those cases, the comparison *properties*—the aspects along which different studies can be compared—must first be identified or designed before a reusable comparison template can be created.

These ideas are further extended in the SmartReviews framework by Oelen *et al.* (2021) [46]. A SmartReview is not a new type of comparison; rather, it is a way of packaging a traditional review article together with its underlying ORKG comparisons and the visualizations generated from them. In other words, a SmartReview bundles (i) ordinary prose, (ii) one or more ORKG comparisons, and (iii) figures that are driven directly by the structured data. This makes the review simultaneously readable as a normal scholarly article and usable as a small, curated data resource. For the ALD community, this simply means that tables such as the rare-earth MOSLED performance table or the SiO_2 ALE summaries are no longer isolated: they can be explored interactively, reused in new plots, or updated as new papers appear.

In the remainder of this article, we adopt these ORKG concepts to encode ALD/E review tables on metal oxides and SiO_2 as comparisons, define reusable property templates for process parameters and mechanisms, and assemble them into SmartReview-style artefacts that support cross-table querying and long-term reuse in the ALD/E community.

3 Machine-actionable ALD/E Review Tables and Q&A in the ORKG

In this section, we demonstrate how review tables from the ALD/E literature can be modeled as machine-actionable comparisons in the Open Research Knowledge Graph (ORKG). Once a table is represented in this structured form, its entries can be retrieved, filtered, and combined using declarative queries rather than by manually scanning PDF documents. To illustrate this in practice, we encode seven ALD tables across three recent review papers and eleven ALE tables across six review papers as ORKG Comparisons. We then formulate a series of natural-language questions about their contents; for each question, we outline the corresponding query and show the complete or partial result. The goal is not to draw new scientific conclusions, but to demonstrate that typical ALD/E review content can be queried deterministically, ensuring factual accuracy and grounding of the retrieved measures once expressed as machine-actionable data.

Atomic layer deposition (ALD) has been extensively reviewed, from the early field-defining article by Suntola [59] and subsequent foundational overviews of precursors and surface chemistry [37, 54] through the comprehensive Kirk–Othmer Encyclopedia chapter by van Ommen, Goulas, and Puurunen [60] and the widely cited field-wide summary by George [22]. Process-focused surveys